

Changin Oh

PhD Candidate | Mathematical Oncology, Computational Biology, Scientific Machine Learning
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Education

Toronto Metropolitan University

Toronto, ON

PhD in Mathematics

2023 – Present

- Thesis: “Reliable Mathematical Modelling in Oncology: Model Formulation, Parameter Estimation, and Data-Driven Discovery”
- Supervisor: Kathleen Wilkie

York University

North York, ON

MA in Mathematics

2021 – 2023

- Thesis: “Linear Spectral Unmixing Algorithms for Abundance Fraction Estimation in Spectroscopy”
- Supervisor: Iain Moyles

University of Western Ontario

London, ON

BSc in Mathematics

2009 – 2017

- Minor in Japanese Studies
- Dean’s Honour List for 2009 – 2011

Publications

Peer-Reviewed Articles

1. M. Oh, **C. Oh**, T. Ghosh, and E. Tabassum, “Selective effects of backbone cyclization and disulfide bonding as global covalent constraints on the conformational ensemble of sunflower trypsin inhibitor-1,” *The Journal of Physical Chemistry B*, Jun 2026.
2. **C. Oh** and K. P. Wilkie, “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection,” *Mathematical Biosciences*, vol. 396, p. 109685, Jun 2026.
- *Featured in Mathematical Oncology Newsletter (research highlight)*
3. M. Oh, **C. Oh**, and E. Tabassum, “Covalent: Interpretable and discriminative collective variables reveal ligand-dependent switching in human cellular retinol-binding protein 2,” *Journal of Chemical Theory and Computation*, vol. 21, no. 23, pp. 12356–12370, Dec 2025.

4. M. Oh, **C. Oh**, and D. F. Weaver, “Effect of cholesterol on the structure of networked water at the surface of a model lipid membrane,” *The Journal of Physical Chemistry B*, vol. 124, no. 18, pp. 3686–3694, May 2020.
5. M. Oh, **C. Oh**, M. Gupta, and D. F. Weaver, “Decoding interfacial water orientation to predict surface charge density on a model sheet using a deep learning algorithm,” *The Journal of Physical Chemistry C*, vol. 124, no. 4, pp. 2574–2582, Jan 2020.
6. M. Oh, M. Gupta, **C. Oh**, and D. F. Weaver, “Understanding the effect of nanoconfinement on the structure of water hydrogen bond networks,” *Physical Chemistry Chemical Physics*, vol. 21, no. 47, pp. 26237–26250, Dec 2019.

Blog Posts

1. A. Köhn-Luque, J. Nardini, **C. Oh**, L. Podina, G. Powathil, E. Rutter, J. West, and K. Wilkie, “Learning dynamics with neural networks,” *Mathematical Oncology Blog*, Mar 2025.

Awards and Prizes

Ontario Graduate Scholarship	2026 – 2027
<i>Awarded as a prestigious, competitive provincial scholarship to graduate students in Ontario</i>	
	2024 – 2025
Toronto Metropolitan Graduate Fellowship	2025 – 2026
<i>Awarded as a competitive, merit-based institutional fellowship to graduate students demonstrating academic excellence</i>	
	2023 – 2024
Mathematics Graduate Award	2023 – 2026
<i>Awarded by the department for outstanding academic performance in graduate mathematics</i>	
Professional Development Grant	2025 – 2026
<i>Awarded by the department for academic dedication and research potential</i>	
Toronto Metropolitan Graduate Student Travel Fund Award	2025
<i>Awarded by the institution to support research dissemination and active participation at academic conferences</i>	
Toronto Metropolitan Graduate Scholarship (declined)	2024 – 2025
<i>Selected as a competitive institutional scholarship recipient through the Ontario Graduate Scholarship competition</i>	
Toronto Metropolitan Graduate Development Award	2024 – 2025
<i>Awarded by the department for academic potential and professional success</i>	
York Graduate Fellowship	2021 – 2022
<i>Awarded as an institutional fellowship supporting academic and research activities</i>	
Classical Studies Prize in Elementary Latin	2016
<i>Awarded by the department to the student with the highest standing in Latin 1000</i>	
Class of '49 Prize	2010
<i>Awarded by the department to the student with the highest mark in Calculus 1501</i>	
Western Scholarship of Excellence	2009
<i>Recognized outstanding high school academic achievement as a merit-based entrance scholarship</i>	

Honours and Distinctions

Featured Research Article

2026/05

Mathematical Oncology Newsletter

- The article “How mathematical forms of chemotherapy and radiotherapy bias model-optimized predictions: Implications for model selection” was featured in *This Week in Mathematical Oncology*, Issue 370.

Presentations

A Gentle Introduction to Riemannian Gradient Descent Method on Smooth Manifolds

- *Chemistry Research Group Meeting (Invited Mini-lecture – Memorial University of Newfoundland)* 2026/01
- *Mathematical Biology Seminar (Mini-lecture – Toronto Metropolitan University)* 2025/10

How Mathematical Forms Describing Cytotoxic Action of Chemotherapy and Radiotherapy Bias Predictions

- *2025 Complex Systems Day (Talk – Toronto)* 2025/08
- *BIRS Workshop on Mechanistic Learning as a Combination of Machine Learning and Modelling in Mathematical Oncology (Poster – Banff)* 2025/01
- *Fields Institute Workshop on Mathematical Modelling of Cancer Treatments, Resistance, Optimization (Poster – Toronto)* 2024/09

Research Experience

Research Assistant

2019/07 – 2020/06

Krembil Research Institute

Toronto, ON

- Topics: (1) Topological analysis on hydrogen bond networks of confined and interfacial waters
(2) Implementation of a deep learning algorithm in indirect surface characterization
(3) Effect of cholesterol on the structure of water at the surface of a model lipid membrane

Research Assistant

2015/05 – 2015/08

University of Western Ontario

London, ON

- Topic: Molecular dynamics studies of nanosized aqueous droplets

Teaching Experience

Guest Instructor

2026/01 – 2026/05

Memorial University of Newfoundland

St. John's, NL

- Designed and delivered intuition-focused lectures in general topology for a chemistry research group.
- Emphasized conceptual understanding rather than formal, proof-based understanding.

Teaching Assistant

2023/09 – Present

Toronto Metropolitan University

Toronto, ON

- Led weekly labs guiding students through difficult problems and reviewing course material.
- Contributed to the development of exam marking schemes (MTH 231).

MTH 141 – Linear Algebra

MTH 207 – Calculus and Computational Methods I

MTH 231 – Modern Mathematics II

MTH 240 – Calculus II

MTH 330 – Calculus and Geometry

MTH 350 – Multivariate and Vector Calculus

MTH 501 – Numerical Analysis I

MTH 430 – Dynamic Systems Differential Equations

MTH 510 – Numerical Analysis

MTH 719 – Applied Linear Algebra

Teaching Assistant

2021/09 – 2022/04

York University

North York, ON

- Provided one-on-one and group tutoring to students

MATH 1013 – Applied Calculus I

MATH 1300 – Differential Calculus with Applications

MATH 1025 – Applied Linear Algebra

MATH 1507 – Mathematics II

Teaching Assistant

2013/09 – 2014/04

The University of Western Ontario

London, ON

- Led group conversation tutorials to support students in improving their Japanese speaking skills.
- Provided instructional support and supervision.

JAPANESE 1036 – Japanese for Beginners

JAPANESE 2260 – Intermediate Japanese

Educational Outreach

Instructor

2026/02 – 2026/06

Toronto Garden Church

North York, ON

- Taught mathematics to students in Grades 5–10 as part of a community outreach program.

Instructor 2025/03 – 2025/06
Toronto Milal Church Toronto, ON

- Taught essential computer skills to elderly learners as part of the senior computer literacy program, including the use of Microsoft Word, Paint, and introductory AI tools.

Instructor 2021/08 – 2024/07
Online Learning Community Korea

- Taught linear algebra, calculus, probability theory, and real analysis online to learners without strong mathematical backgrounds.
- YouTube Channel: [Math Friends](#)

Instructor 2015/09 – 2016/08
London Full Gospel Church London, ON

- Taught mathematics and Japanese to secondary school students as part of a community outreach program.

Instructor 2014/09 – 2015/08
Oakridge Presbyterian Church London, ON

- Taught Grade 9–12 mathematics to secondary school students as part of a community outreach program.

Instructor 2008/05 – 2009/07
Oakville Presbyterian Church Oakville, ON

- Taught mathematics and Japanese to immigrant secondary school students as part of a community outreach program.

Technical and Language Skills

- Programming Languages
 - Julia
 - MATLAB
 - Python
- Languages
 - Korean – Native
 - English – Advanced
 - Japanese – Advanced (JLPT N1)